



115 年淹水數值模型第三次更新輔助管理  
教育訓練

# Delft3D FM 1D2D

## 資料轉換工具

國立臺灣海洋大學

2026/05/29 (五)



經濟部水利署水利規劃分署  
國立臺灣海洋大學  
國立成功大學  
國立臺北科技大學  
方圖科技股份有限公司  
三川雲科技股份有限公司  
峰昀有限公司



- 環境設置
  - 確認 Miniforge Prompt 設定
  - 升級 d3dttools 版本
- d3dttools工具
  - 資料轉換
  - 降雨資料
  - 網格移除
  - IOT驗證

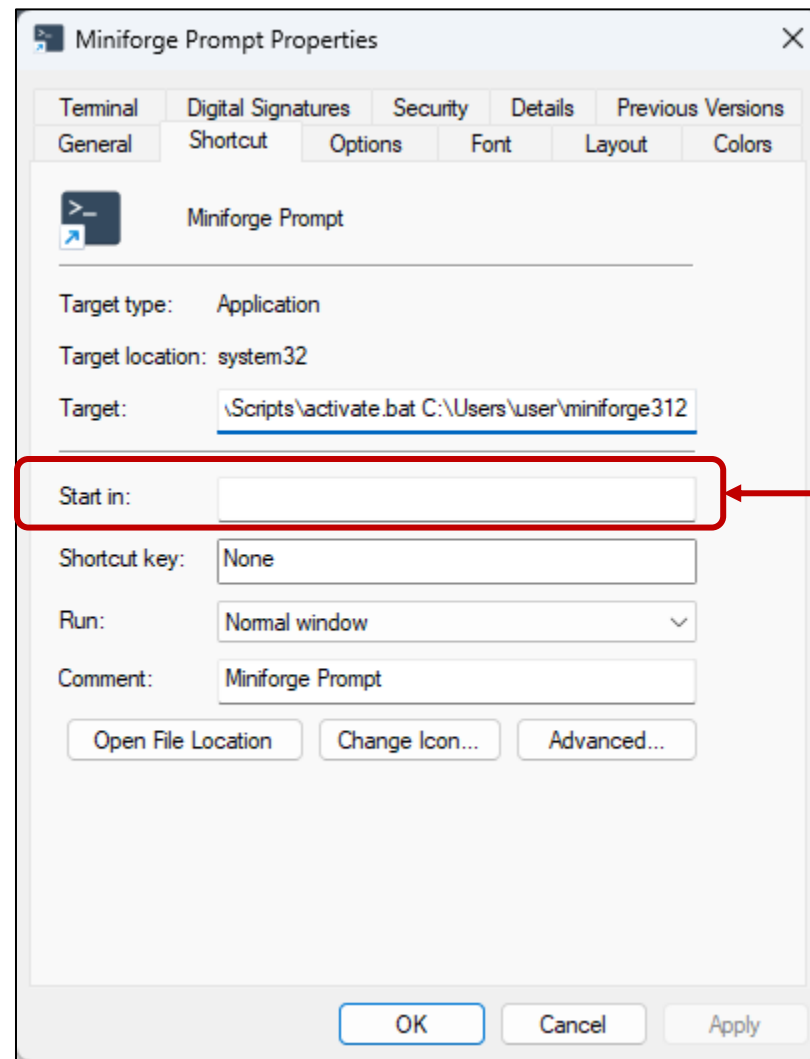
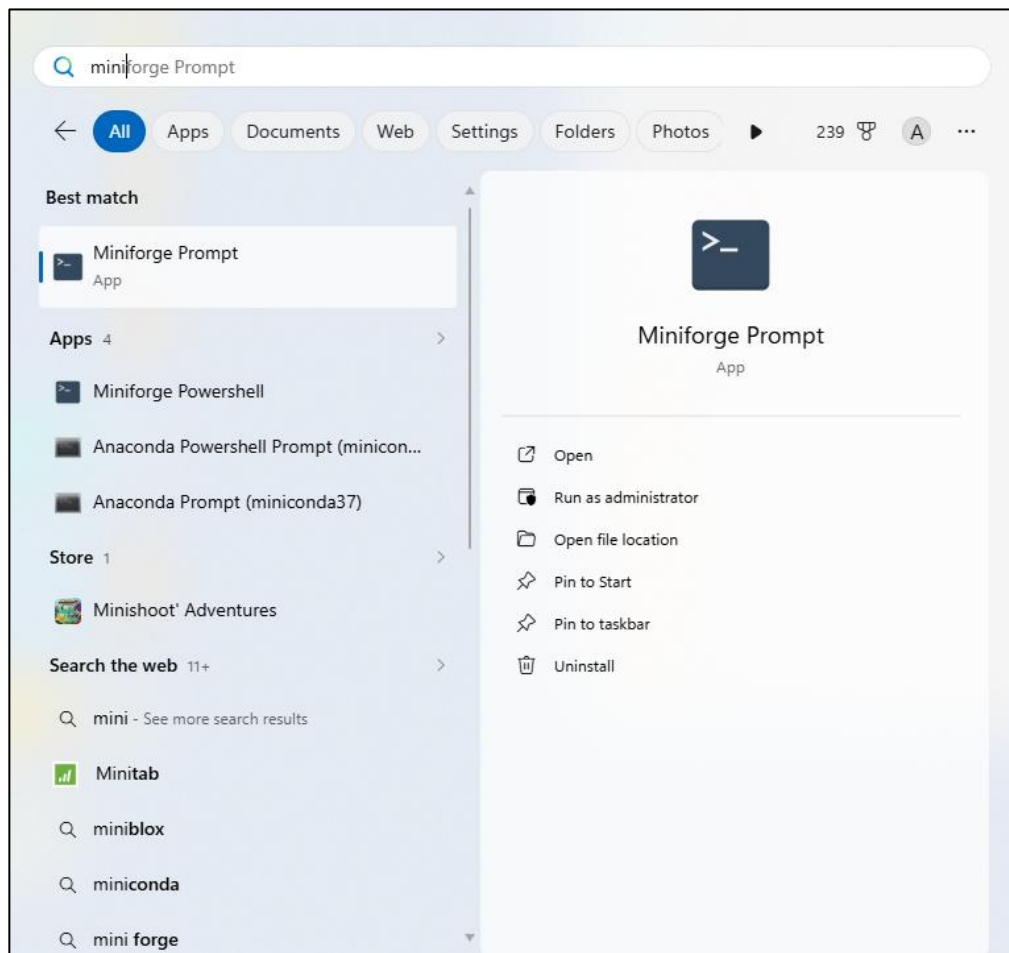


# 環境設置

# Python 環境設置



## 確認 Miniforge Prompt 設定



需清空



## 升級 d3dttools 版本

```
Miniforge Prompt - conda de x + v
(base) C:\Users\user\Desktop>conda activate d3dtoolsevn
(d3dtoolsevn) C:\Users\user\Desktop>pip install d3dtools --force-reinstall
```

啟動環境

強制升級版本



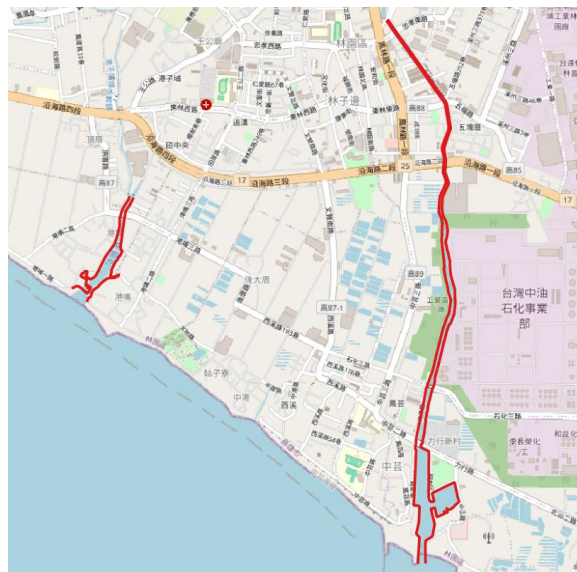
# d3dtools

## Part1-資料轉換

## 資料轉換

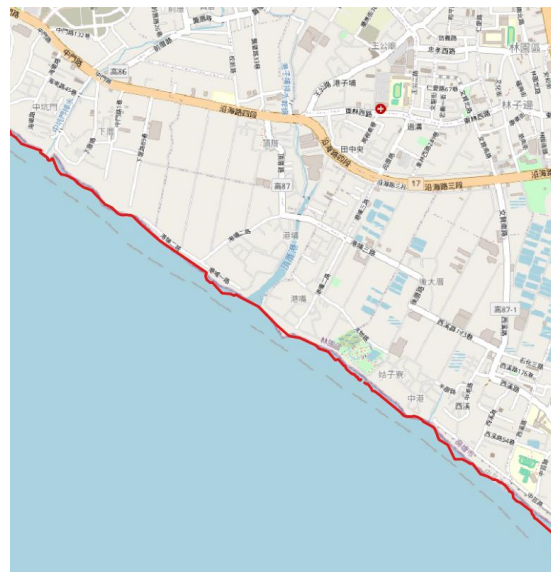
情境

溢堤線



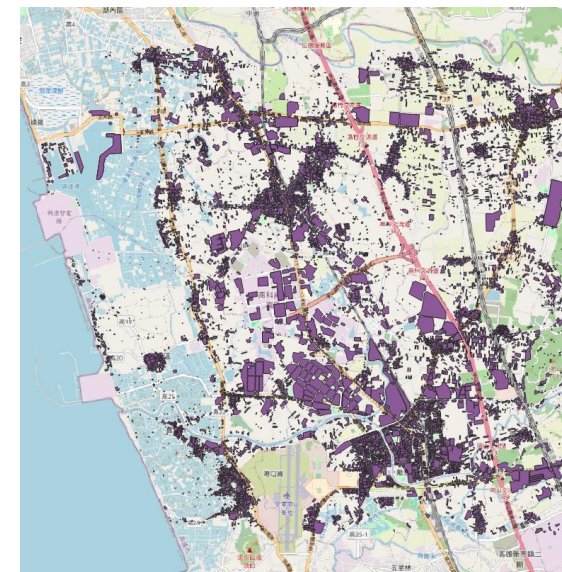
shapefile → \*.pliz

邊界線



shapefile → \*.pli

建物區塊



shapefile → \*.pol

檔案

| Name                                                                                          |
|-----------------------------------------------------------------------------------------------|
|  SHP_BC    |
|  SHP_BLOCK |
|  SHP_DIKE  |

| Date modified     | Type        | Size |
|-------------------|-------------|------|
| 5/23/2026 4:37 PM | File folder |      |
| 5/23/2026 4:37 PM | File folder |      |
| 5/23/2026 4:37 PM | File folder |      |

## 資料轉換

指令

Name

SHP\_BC

SHP\_BLOCK

SHP\_DIKE

Miniforge Prompt

```
Miniforge Prompt
(base) C:\Users\user\Desktop\TUTORIAL\PART1_CNV>conda activate d3dtoolsenv 啟動環境
(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART1_CNV>shp2pli shapefile→*.pli
Found 1 shapefiles in SHP_BC
Created output folder: PLI_BC
Processing: DSBC
Done! Generated 2 PLI files in PLI_BC

(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART1_CNV>shp2pliz shapefile→*.pliz
Found 1 shapefiles in SHP_DIKE
Processing: BANKLINE_LY
Done! PLIZ file created at: PLIZ_DIKE\Dike.pliz

(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART1_CNV>shp2pol shapefile→*.pol
Processing block: BLOCK_AG
Conversion complete: 1 shapefiles processed

(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART1_CNV>
```



## 資料轉換

結果

生成檔案

| Name             | Date modified     | Type        | Size |
|------------------|-------------------|-------------|------|
| PLI_BC           | 5/23/2026 8:20 PM | File folder |      |
| PLIZ_DIKE        | 5/23/2026 8:20 PM | File folder |      |
| POL_BLOCK        | 5/23/2026 8:20 PM | File folder |      |
| SHP_BC           | 5/23/2026 4:37 PM | File folder |      |
| SHP_BLOCK        | 5/23/2026 4:37 PM | File folder |      |
| SHP_DIKE         | 5/23/2026 4:37 PM | File folder |      |
| Miniforge Prompt | 5/22/2026 8:36 PM | Shortcut    |      |



# d3dtools

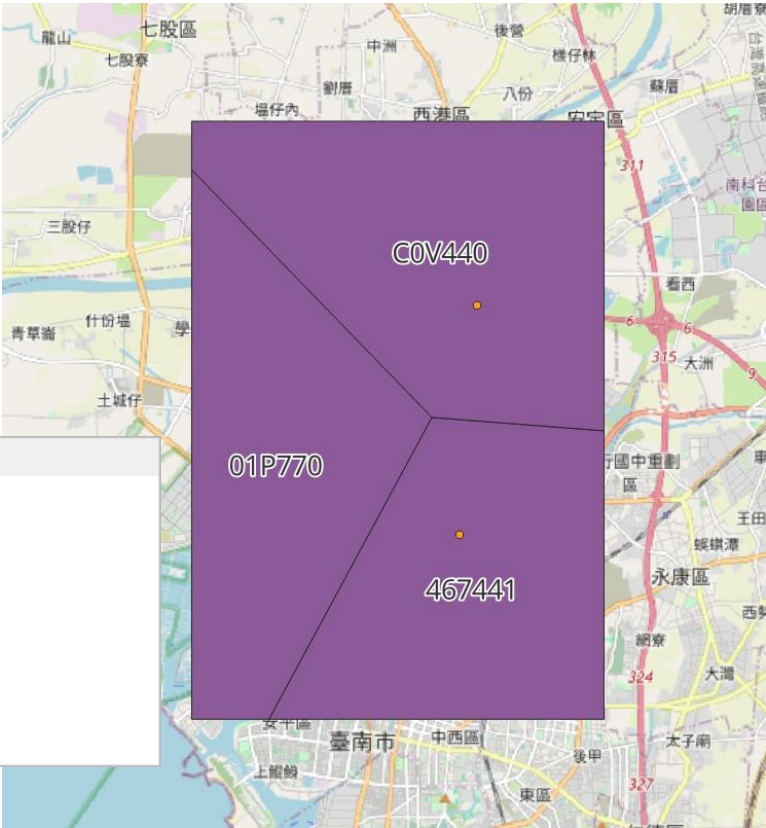
## Part2-降雨資料



## 降雨資料 情境降雨

情境

|   | id | Station |
|---|----|---------|
| 1 | 1  | 01P770  |
| 2 | 3  | 467441  |
| 3 | 2  | COV440  |



| kaohsiung6hr.csv > data |       |           |           |           |     |     |       |       |        |        |       |       |       |
|-------------------------|-------|-----------|-----------|-----------|-----|-----|-------|-------|--------|--------|-------|-------|-------|
| ID                      | 站名    | Horner(a) | Horner(b) | Horner(c) | 情境  | 0小時 | 1小時   | 2小時   | 3小時    | 4小時    | 5小時   | 6小時   | 合計百分比 |
| 467441                  | 高雄    | 3823.735  | 29.198    | 0.672     | 100 | 0   | 7.84  | 12.4  | 44.85  | 18.64  | 9.52  | 6.75  | 100   |
| 467441                  | 高雄    | 3823.735  | 29.198    | 0.672     | 150 | 0   | 11.76 | 18.6  | 67.28  | 27.96  | 14.28 | 10.13 | 100   |
| 467441                  | 高雄    | 3823.735  | 29.198    | 0.672     | 200 | 0   | 15.68 | 24.8  | 89.7   | 37.28  | 19.04 | 13.5  | 100   |
| 467441                  | 高雄    | 3823.735  | 29.198    | 0.672     | 250 | 0   | 19.6  | 31    | 112.13 | 46.6   | 23.8  | 16.88 | 100   |
| 467441                  | 高雄    | 3823.735  | 29.198    | 0.672     | 300 | 0   | 23.52 | 37.2  | 134.55 | 55.92  | 28.56 | 20.25 | 100   |
| 467441                  | 高雄    | 3823.735  | 29.198    | 0.672     | 350 | 0   | 27.44 | 43.4  | 156.98 | 65.24  | 33.32 | 23.63 | 100   |
| 467441                  | 高雄    | 3823.735  | 29.198    | 0.672     | 400 | 0   | 31.36 | 49.6  | 179.4  | 74.56  | 38.08 | 27    | 100   |
| 467441                  | 高雄    | 3823.735  | 29.198    | 0.672     | 450 | 0   | 35.28 | 55.8  | 201.83 | 83.88  | 42.84 | 30.38 | 100   |
| 467441                  | 高雄    | 3823.735  | 29.198    | 0.672     | 500 | 0   | 39.2  | 62    | 224.25 | 93.2   | 47.6  | 33.75 | 100   |
| COV440                  | 鳳山    | 24141.994 | 390.914   | 0.868     | 100 | 0   | 12.02 | 16.87 | 25.95  | 20.61  | 14.12 | 10.43 | 100   |
| COV440                  | 鳳山    | 24141.994 | 390.914   | 0.868     | 150 | 0   | 18.03 | 25.31 | 38.93  | 30.92  | 21.18 | 15.65 | 100   |
| COV440                  | 鳳山    | 24141.994 | 390.914   | 0.868     | 200 | 0   | 24.04 | 33.74 | 51.9   | 41.22  | 28.24 | 20.86 | 100   |
| COV440                  | 鳳山    | 24141.994 | 390.914   | 0.868     | 250 | 0   | 30.05 | 42.18 | 64.88  | 51.53  | 35.3  | 26.08 | 100   |
| COV440                  | 鳳山    | 24141.994 | 390.914   | 0.868     | 300 | 0   | 36.06 | 50.61 | 77.85  | 61.83  | 42.36 | 31.29 | 100   |
| COV440                  | 鳳山    | 24141.994 | 390.914   | 0.868     | 350 | 0   | 42.07 | 59.05 | 90.83  | 72.14  | 49.42 | 36.51 | 100   |
| COV440                  | 鳳山    | 24141.994 | 390.914   | 0.868     | 400 | 0   | 48.08 | 67.48 | 103.8  | 82.44  | 56.48 | 41.72 | 100   |
| COV440                  | 鳳山    | 24141.994 | 390.914   | 0.868     | 450 | 0   | 54.09 | 75.92 | 116.78 | 92.75  | 63.54 | 46.94 | 100   |
| COV440                  | 鳳山    | 24141.994 | 390.914   | 0.868     | 500 | 0   | 60.1  | 84.35 | 129.75 | 103.05 | 70.6  | 52.15 | 100   |
| 01P770                  | 美濃(2) | 1488.414  | 50.842    | 0.548     | 100 | 0   | 10.3  | 14.67 | 34.17  | 19.74  | 11.98 | 9.14  | 100   |
| 01P770                  | 美濃(2) | 1488.414  | 50.842    | 0.548     | 150 | 0   | 15.45 | 22.01 | 51.26  | 29.61  | 17.97 | 13.71 | 100   |
| 01P770                  | 美濃(2) | 1488.414  | 50.842    | 0.548     | 200 | 0   | 20.6  | 29.34 | 68.34  | 39.48  | 23.96 | 18.28 | 100   |
| 01P770                  | 美濃(2) | 1488.414  | 50.842    | 0.548     | 250 | 0   | 25.75 | 36.68 | 85.43  | 49.35  | 29.95 | 22.85 | 100   |
| 01P770                  | 美濃(2) | 1488.414  | 50.842    | 0.548     | 300 | 0   | 30.9  | 44.01 | 102.51 | 59.22  | 35.94 | 27.42 | 100   |
| 01P770                  | 美濃(2) | 1488.414  | 50.842    | 0.548     | 350 | 0   | 36.05 | 51.35 | 119.6  | 69.09  | 41.93 | 31.99 | 100   |
| 01P770                  | 美濃(2) | 1488.414  | 50.842    | 0.548     | 400 | 0   | 41.2  | 58.68 | 136.68 | 78.96  | 47.92 | 36.56 | 100   |
| 01P770                  | 美濃(2) | 1488.414  | 50.842    | 0.548     | 450 | 0   | 46.35 | 66.02 | 153.77 | 88.83  | 53.91 | 41.13 | 100   |
| 01P770                  | 美濃(2) | 1488.414  | 50.842    | 0.548     | 500 | 0   | 51.5  | 73.35 | 170.85 | 98.7   | 59.9  | 45.7  | 100   |

檔案

| Name             | Date modified     | Type                 |
|------------------|-------------------|----------------------|
| SHP              | 5/23/2026 4:30 PM | File folder          |
| kaohsiung6hr.csv | 5/16/2025 9:46 AM | Microsoft Excel C... |

## 降雨資料

情境降雨

指令

「情境降雨」資料夾

Name

SHP

kaohsiung6hr.csv

Miniforge Prompt

結果

Name

SHP

TAB

kaohsiung6hr.csv

Miniforge Prompt

```
Miniforge Prompt
(base) C:\Users\user\Desktop\TUTORIAL\PART2_NC\情境降雨>conda activate d3dtoolsenv
(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART2_NC\情境降雨>snorain -i kaohsiung6hr.csv
```

啟動環境

生成各情境時間序列

生成檔案

```
C: > Users > user > Desktop > TUTORIAL > PART2_NC > 情境降雨 > TAB > kaohsiung6hr_100.csv > data
1 time,01P770,467441,C0V440
2 2000-01-01 00:00:00,0.0,0.0,0.0
3 2000-01-01 01:00:00,10.3,7.84,12.02
4 2000-01-01 02:00:00,14.67,12.4,16.87
5 2000-01-01 03:00:00,34.17,44.85,25.95
6 2000-01-01 04:00:00,19.74,18.64,20.61
7 2000-01-01 05:00:00,11.98,9.52,14.12
8 2000-01-01 06:00:00,9.14,6.75,10.43
9
```

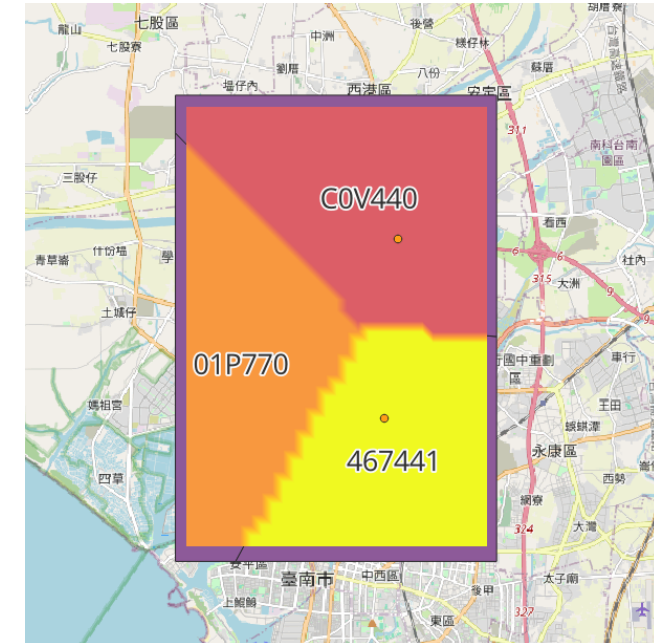
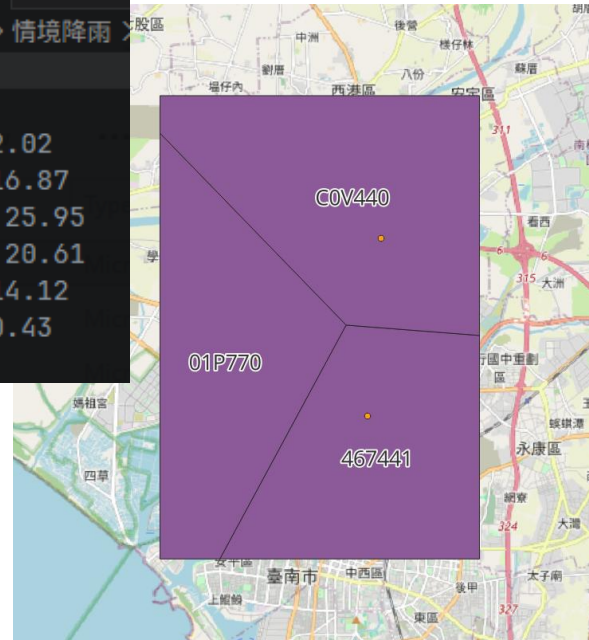
## 降雨資料

情境降雨

情境

C: > Users > user > Desktop > TUTORIAL > PART2\_NC > 情境降雨

```
1 time,01P770,467441,C0V440
2 2000-01-01 00:00:00,0.0,0.0,0.0
3 2000-01-01 01:00:00,10.3,7.84,12.02
4 2000-01-01 02:00:00,14.67,12.4,16.87
5 2000-01-01 03:00:00,34.17,44.85,25.95
6 2000-01-01 04:00:00,19.74,18.64,20.61
7 2000-01-01 05:00:00,11.98,9.52,14.12
8 2000-01-01 06:00:00,9.14,6.75,10.43
9
```



netcdf 降雨

檔案

Name

SHP

TAB

kaohsiung6hr.csv

Miniforge Prompt

Name

kaohsiung6hr\_100.csv

kaohsiung6hr\_150.csv

kaohsiung6hr\_200.csv

kaohsiung6hr\_250.csv

kaohsiung6hr\_300.csv

kaohsiung6hr\_350.csv

kaohsiung6hr\_400.csv

kaohsiung6hr\_450.csv

kaohsiung6hr\_500.csv

## 降雨資料

情境降雨

指令

「情境降雨」資料夾

Name

SHR

TAB

kaohsiung6hr.csv

Miniforge Prompt

結果

Name

NC

SHR

TAB

kaohsiung6hr.csv

Miniforge Prompt

```
Miniforge Prompt
(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART2_NC\情境降雨>ncrain
(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART2_NC\情境降雨>
```

生成各情境  
netcdf 降雨

Name

kaohsiung6hr\_100.nc

kaohsiung6hr\_150.nc

kaohsiung6hr\_200.nc

kaohsiung6hr\_250.nc

kaohsiung6hr\_300.nc

kaohsiung6hr\_350.nc

kaohsiung6hr\_400.nc

kaohsiung6hr\_450.nc

kaohsiung6hr\_500.nc

生成檔案

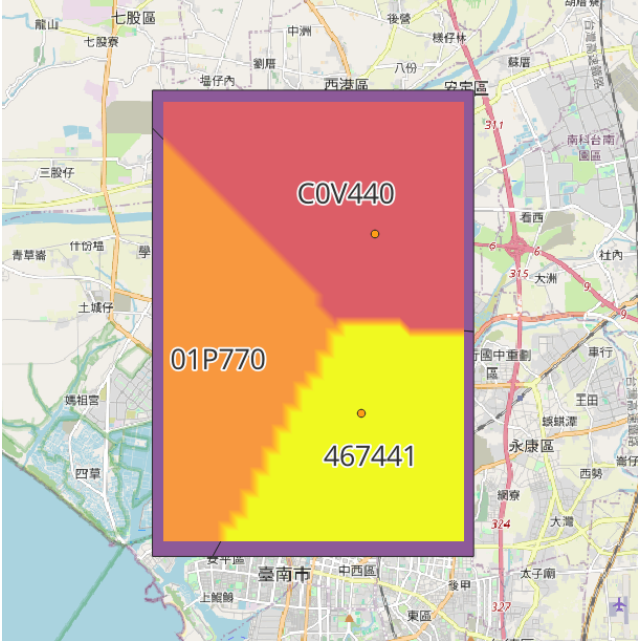
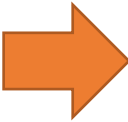
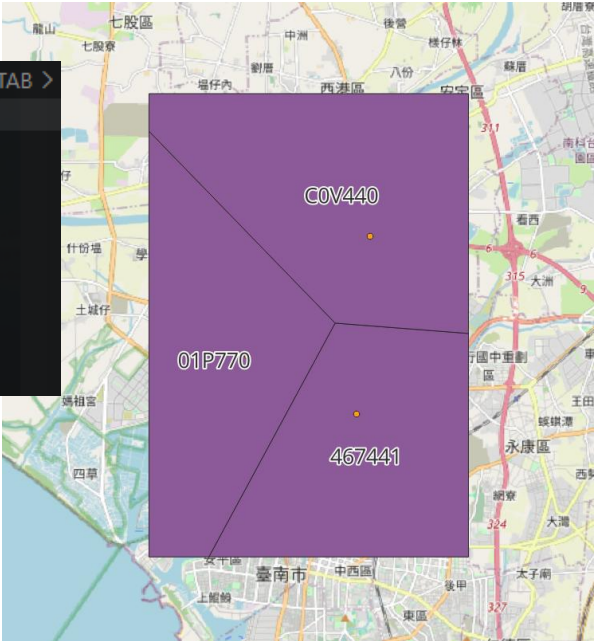


## 降雨資料

事件降雨

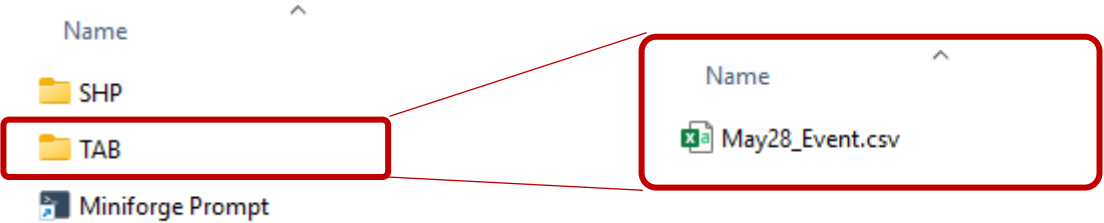
情境

```
C: > Users > user > Desktop > TUTORIAL > PART2_NC > 事件降雨 > TAB >  
1 time,01P770,467441,C0V440  
2 2026-05-28 01:00:00,0.0,0.0,0.0  
3 2026-05-28 02:00:00,51.5,39.2,60.1  
4 2026-05-28 03:00:00,73.35,62.0,84.35  
5 2026-05-28 04:00:00,170.85,224.25,129.75  
6 2026-05-28 05:00:00,98.7,93.2,103.05  
7 2026-05-28 06:00:00,59.9,47.6,70.6  
8 2026-05-28 07:00:00,45.7,33.75,52.15  
9
```



netcdf 降雨

檔案



## 降雨資料

事件降雨

指令

「事件降雨」資料夾

Name

SH

TAB

Miniforge Prompt

結果

Name

NC

SH

TAB

Miniforge Prompt

```
Miniforge Prompt
(base) C:\Users\user\Desktop\TUTORIAL\PART2_NC\事件降雨>conda activate d3dtoolsenv
(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART2_NC\事件降雨>ncrain 生成各事件 netcdf 降雨
```

生成檔案

Name

May28\_Event.nc





# d3dtools

## Part3-網格移除

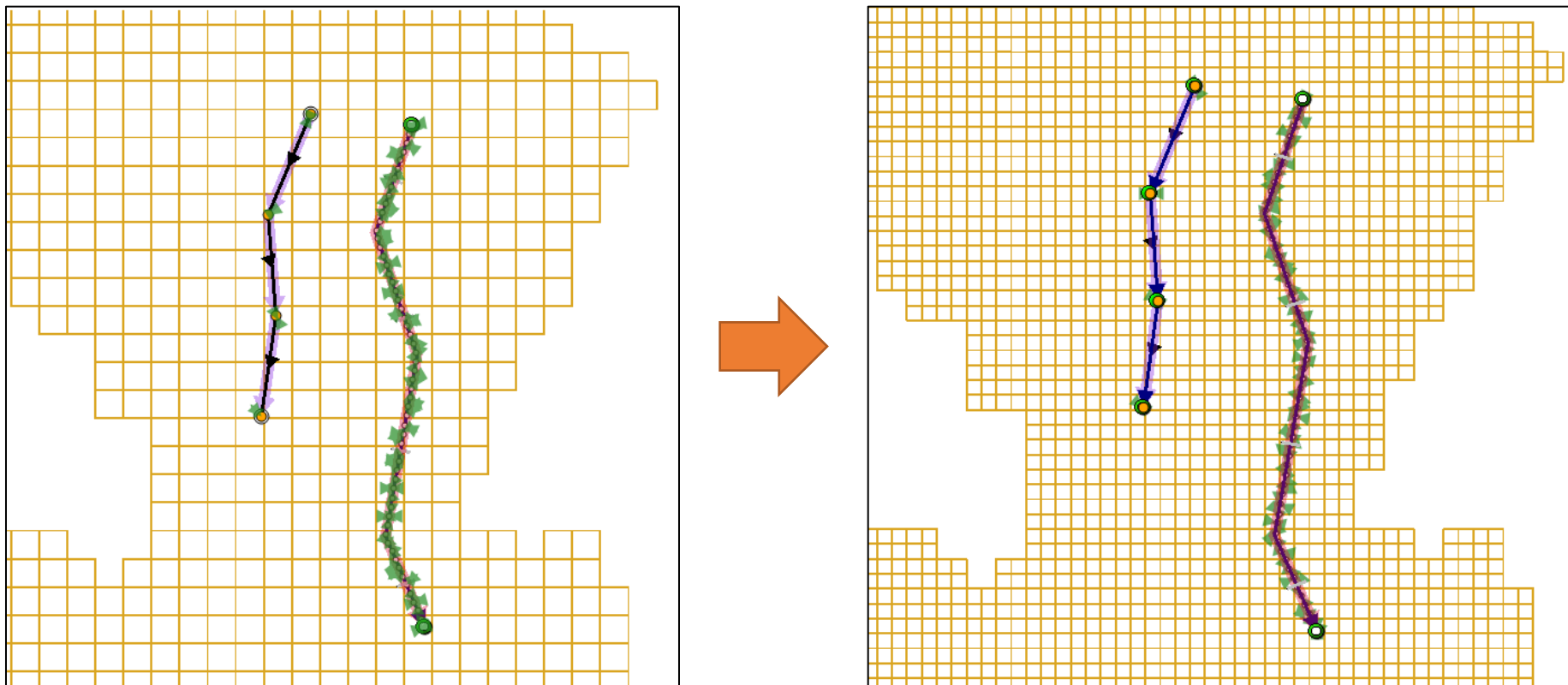
## 網格移除

情境

檔案

Name

- UGRID
- RMGRID.dsproj\_data
- RMGRID.dsproj
- Miniforge Prompt



## 網格移除

### 指令

Name

- UGRID
- RMGRID.dsproj\_data
- RMGRID.dsproj
- Miniforge Prompt

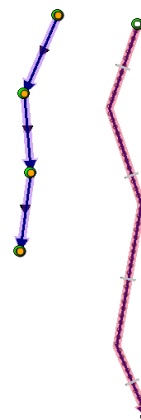
### 結果

Name

- UGRID
- RMGRID.dsproj\_data
- RMGRID.dsproj
- Miniforge Prompt

```
Miniforge Prompt
(base) C:\Users\user\Desktop\TUTORIAL\PART3_GRID>conda activate d3dtoolsenv
(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART3_GRID>rmgrid -i RMGRID.dsproj
```

移除專案網格



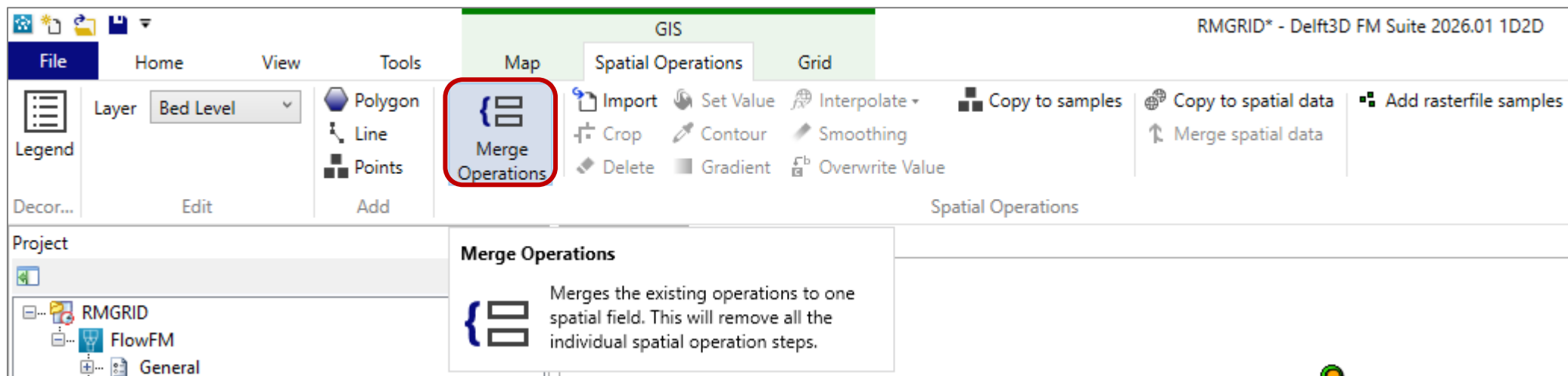
1. 網格移除
2. 1D2D Links 移除
3. Computation nodes 保留

m 200 400 600 800

## 網格移除

### 注意事項

使用 **rmgrid** 前確認執行 **Merge Operations**





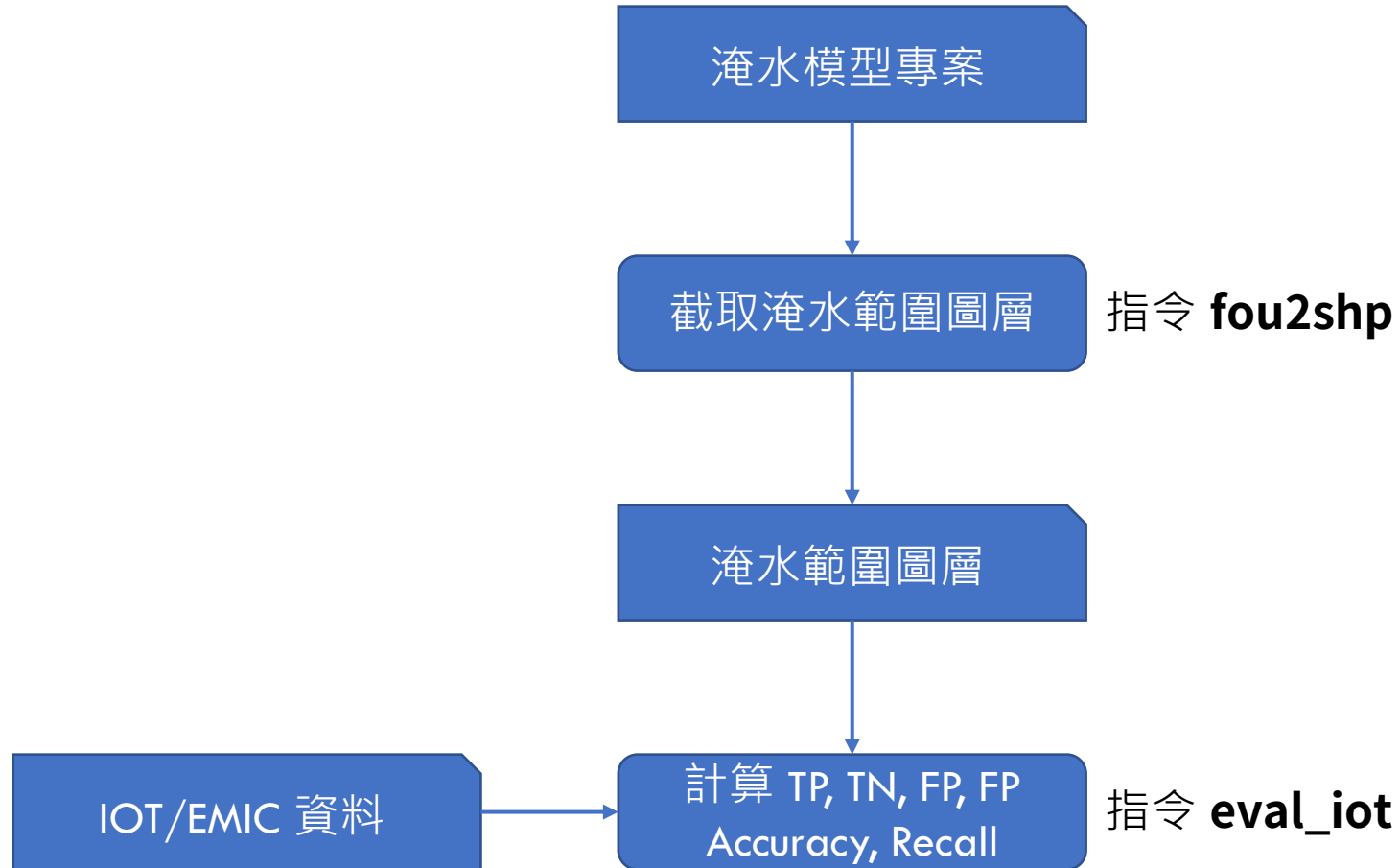
# d3dtools

## Part4-IOT驗證



## IOT驗證

情境



## IOT驗證

### 判斷準則

### 圖示

● 淹水門檻 47.5cm 淹水範圍

● 淹水門檻 30.0cm 淹水範圍

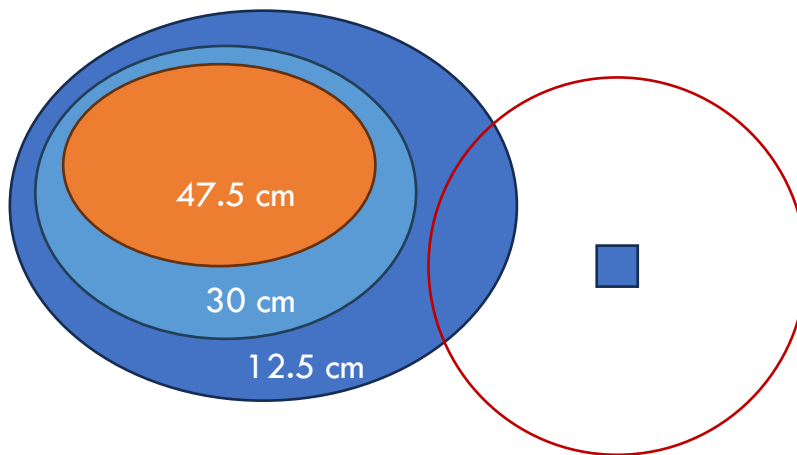
● 淹水門檻 12.5cm 淹水範圍

■ IOT 淹水  $\geq 10$  cm

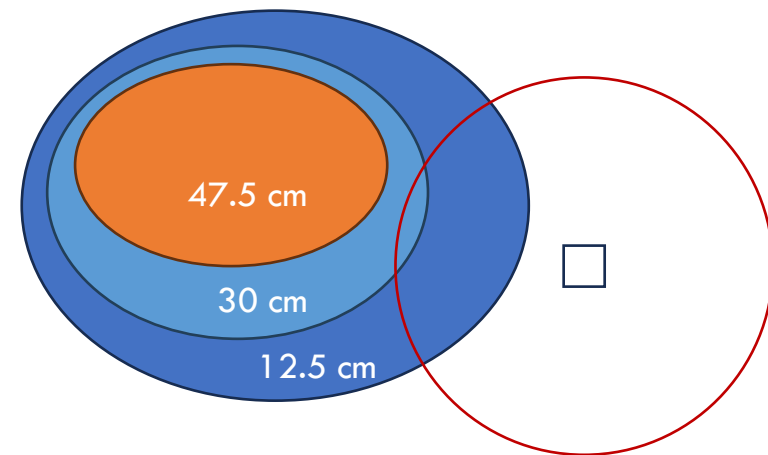
□ IOT 淹水  $< 10$  cm

○ IOT 搜尋半徑

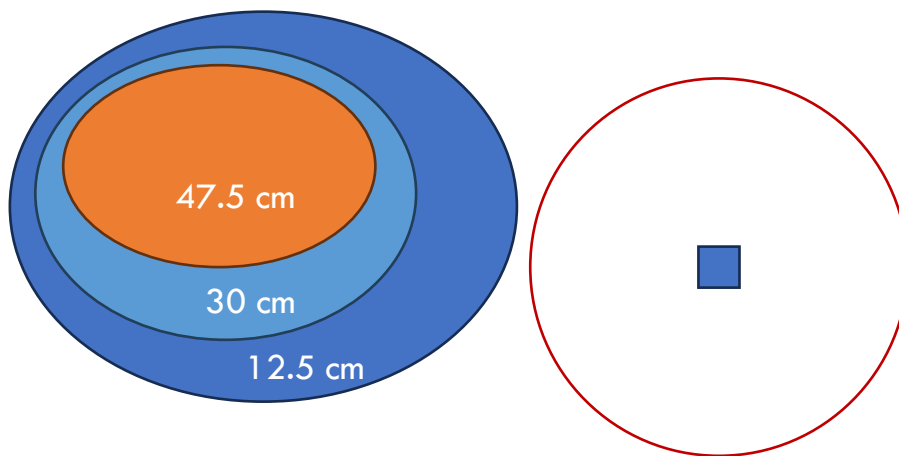
觀測有淹，模擬有淹



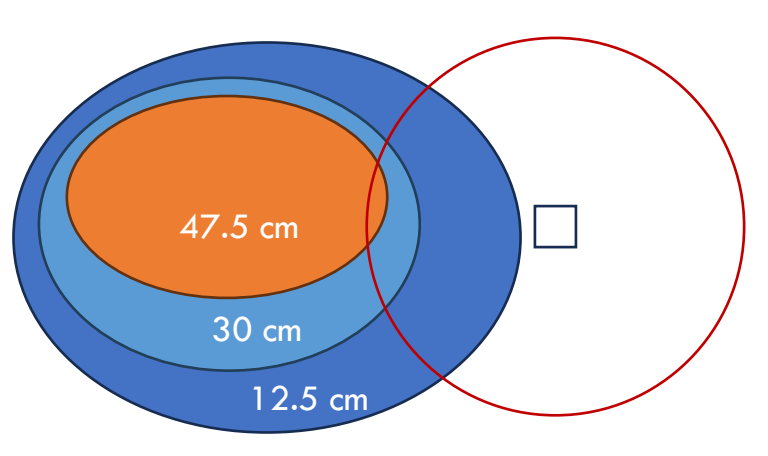
觀測沒淹，模擬沒淹



觀測有淹，模擬沒淹



觀測沒淹，模擬有淹



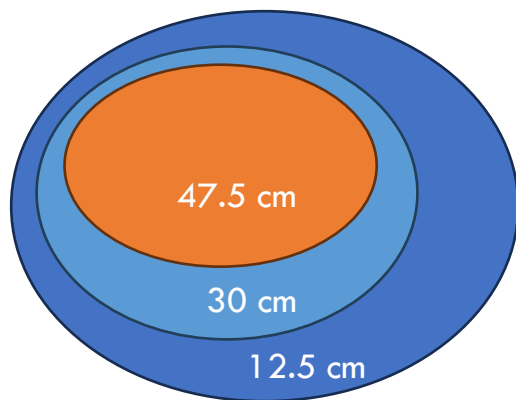


## IOT驗證

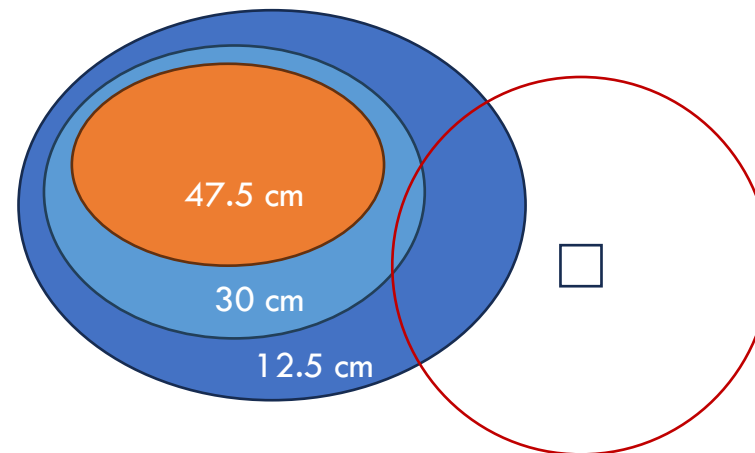
判斷準則

特殊情況

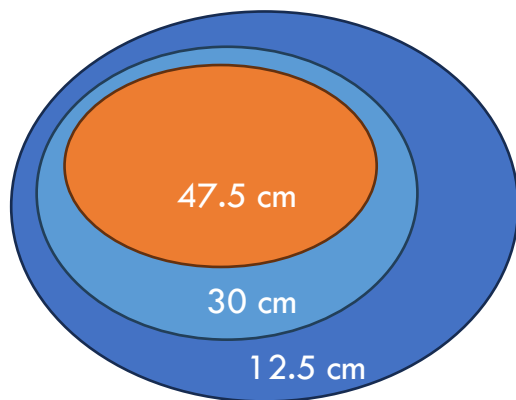
觀測有淹，模擬沒淹



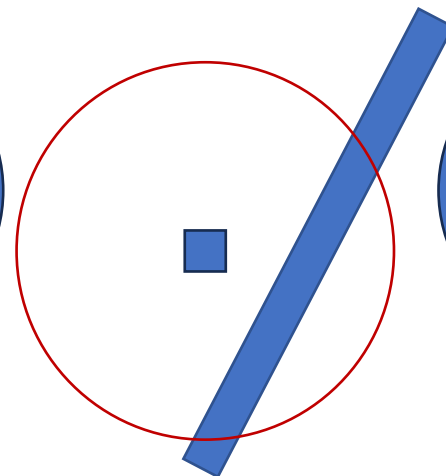
觀測沒淹，模擬沒淹



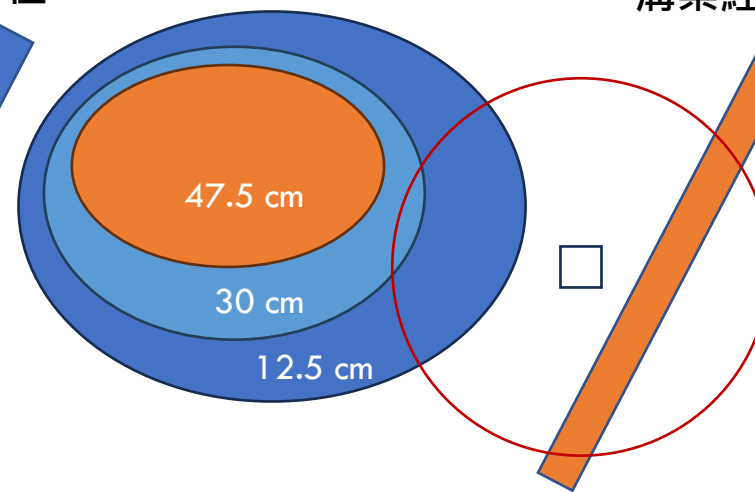
觀測有淹，模擬有淹



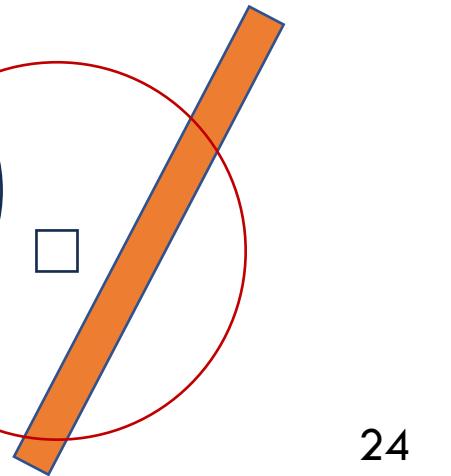
溝渠經過搜尋半徑



觀測沒淹，模擬有淹



溝渠經過搜尋半徑



## 圖示

淹水門檻 47.5cm 淹水範圍

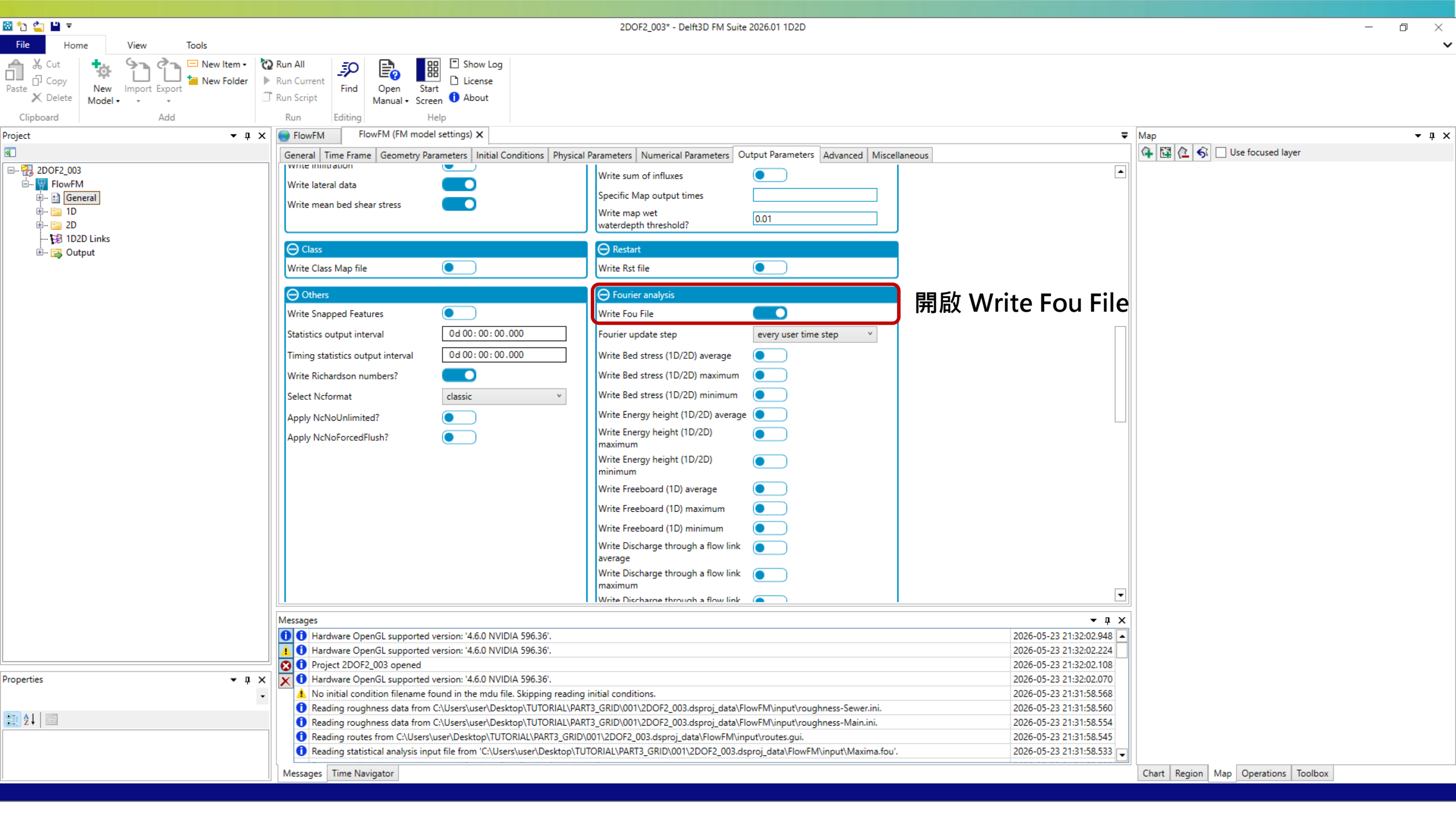
淹水門檻 30.0cm 淹水範圍

淹水門檻 12.5cm 淹水範圍

IOT 淹水  $\geq 10$  cm

IOT 淹水  $< 10$  cm

IOT 搜尋半徑



開啟 Write Fou File

## IOT驗證

### 檔案

Name

2DOF2\_003.dsproj\_data

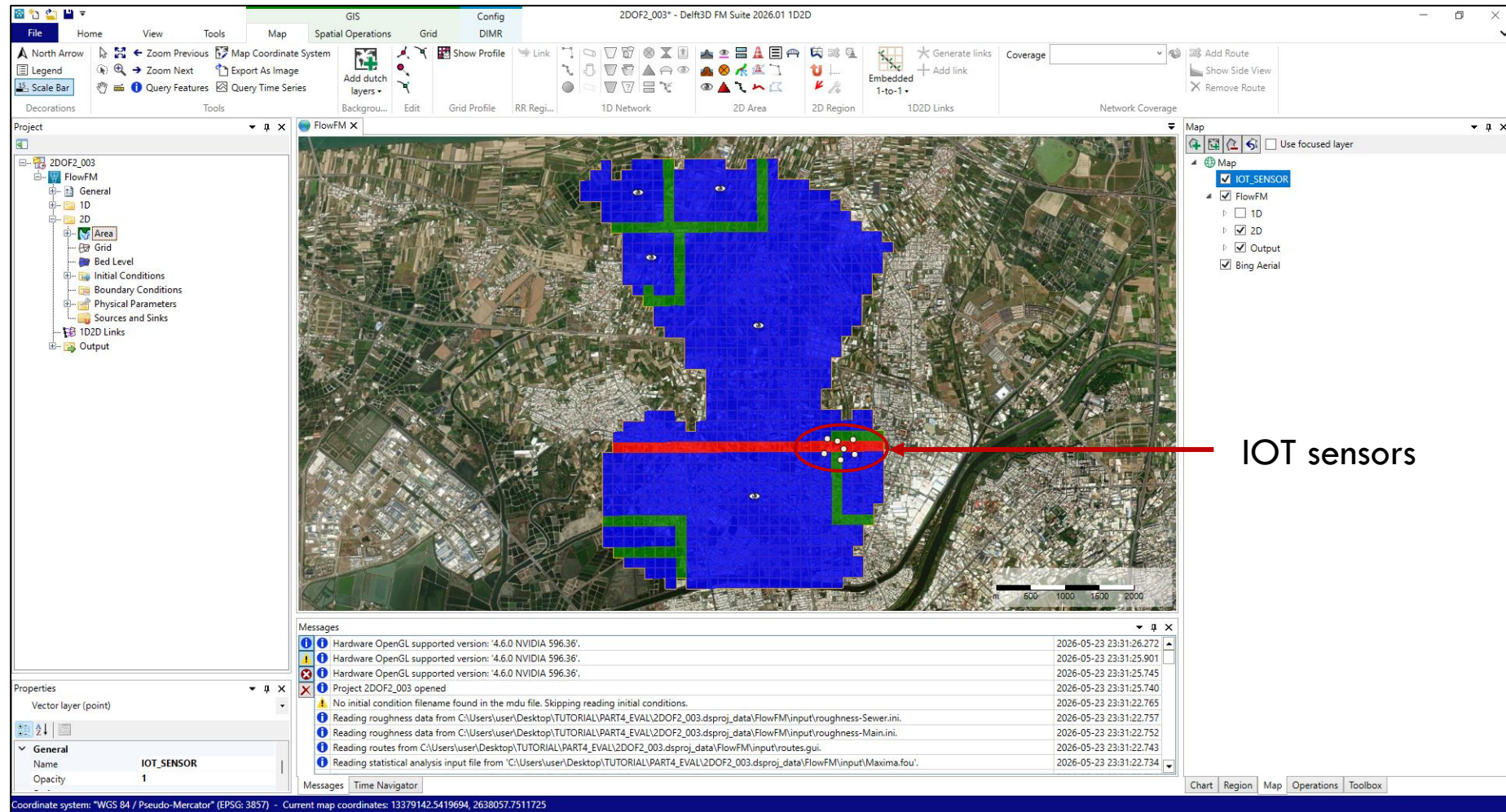
NC

SHP

2DOF2\_003.dsproj

Miniforge Prompt

預設已執行模擬，  
並將 FlowFM/output/FlowFM\_fou.nc  
複製至 NC 資料夾



## IOT驗證

### 指令

- Name ^
- 2DOF2\_003.dsproj\_data
  - NC
  - SHP
  - 2DOF2\_003.dsproj
  - Miniforge Prompt

### 截取淹水範圍圖層

```
Miniforge Prompt
(base) C:\Users\user\Desktop\TUTORIAL\PART4_EVAL>conda activate d3dtoolsenv
(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART4_EVAL>fou2shp -i .\NC\FlowFM_fou.nc
```

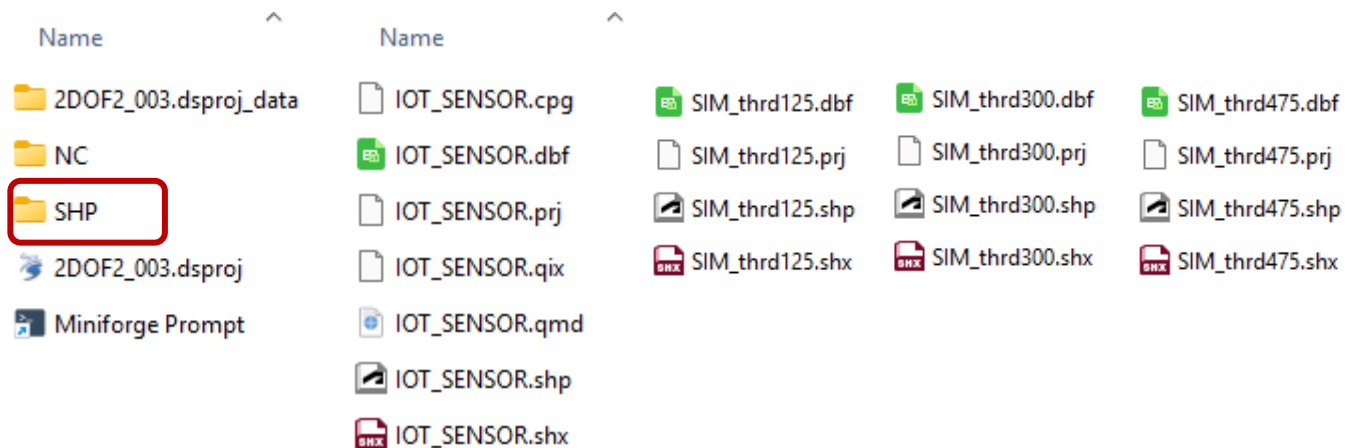
### 結果

- Name ^
- |                       |                 |                 |                 |
|-----------------------|-----------------|-----------------|-----------------|
| 2DOF2_003.dsproj_data |                 |                 |                 |
| NC                    | SIM_thrd125.dbf | SIM_thrd300.dbf | SIM_thrd475.dbf |
| SHP                   | SIM_thrd125.prj | SIM_thrd300.prj | SIM_thrd475.prj |
| 2DOF2_003.dsproj      | SIM_thrd125.shp | SIM_thrd300.shp | SIM_thrd475.shp |
| Miniforge Prompt      | SIM_thrd125.shx | SIM_thrd300.shx | SIM_thrd475.shx |



## IOT驗證

檔案



IOT\_SENSOR — Features Total: 7, Filtered: 7, Selected: 0

| 序號 | 災點分   | 災情來    | 來源說   | 編號   | 縣市             | 鄉鎮  | 災害種 | 發生時 | 災害地       | 災情描       | 最大深       | 是否退 | 退水時 | 災害處       | 詢問單          | 經度                       |
|----|-------|--------|-------|------|----------------|-----|-----|-----|-----------|-----------|-----------|-----|-----|-----------|--------------|--------------------------|
| 1  | 30351 | 經濟部水利署 | 消防署   | EMIC | 42024057019467 | 高雄市 | 鳳山區 | 道路  | 7/25/2024 | 高雄市鳳山區... | 反映該處已淹... | 30  | 已退水 | 7/25/2024 | EMIC資訊:已處... | 聯絡人:聯絡電話: 120.3638370000 |
| 2  | 29989 | 經濟部水利署 | 消防署   | EMIC | 42024057016878 | 高雄市 | 鳳山區 | 道路  | 7/25/2024 | 高雄市鳳山區... | 單行這積水看... | 5   | 已退水 | 7/25/2024 | EMIC資訊:待處... | 聯絡人:聯絡電話: 120.3554359000 |
| 3  | 28863 | 第六河川分署 | 消防署   | EMIC | 42024057009681 | 高雄市 | 鳳山區 | 道路  | 7/25/2024 | 高雄市鳳山區... | 道路淹水      | 35  | 已退水 | 7/25/2024 | EMIC資訊:待處... | 聯絡人:聯絡電話: 120.3554359000 |
| 4  | 28527 | 第六河川分署 | 消防署   | EMIC | 42024057007529 | 高雄市 | 鳳山區 | 道路  | 7/25/2024 | 高雄市鳳山區... | 路面淹水至腳踝   | 6   | 已退水 | 7/25/2024 | EMIC資訊:待處... | 聯絡人:聯絡電話: 120.3554359000 |
| 5  | 28440 | 第六河川分署 | 高雄市政府 | 淹水感測 | 新富路與大明...      | 高雄市 | 鳳山區 | 道路  | 7/25/2024 | 新富路與大明... | 目前感測值0公分  | 33  | 已退水 | 7/25/2024 | NULL         | 聯絡人:聯絡電話: 120.3415400000 |
| 6  | 28181 | 經濟部水利署 | 消防署   | EMIC | 42024056998170 | 高雄市 | 大寮區 | 其他  | 7/24/2024 | 高雄市大寮區... | 光華路側溝涵... | 4   | 已退水 | 7/25/2024 | EMIC資訊:待處... | 聯絡人:聯絡電話: 120.3835290000 |
| 7  | 28440 | 第六河川分署 | 高雄市政府 | 淹水感測 | 新富路與大明...      | 高雄市 | 鳳山區 | 道路  | 7/25/2024 | 新富路與大明... | 目前感測值0公分  | 3   | 已退水 | 7/25/2024 | NULL         | 聯絡人:聯絡電話: 120.3415400000 |

Show All Features



## IOT驗證

### 指令

Name

2DOF2\_003.dsproj\_data

NC

2DOF2\_003.dsproj

Miniforge Prompt

### 結果

計算 TP, TN, FP, FN  
Accuracy, Recall

```
Miniforge Prompt
(base) C:\Users\user\Desktop\TUTORIAL\PART4_EVAL>conda activate d3dtoolsenv
(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART4_EVAL>eval_iot

Miniforge Prompt
(d3dtoolsenv) C:\Users\user\Desktop\TUTORIAL\PART4_EVAL>eval_iot --buffer-by-source EMIC=20 淹水感測=30
```

```
Buffer geometries saved to SHP\IOT_BUFFER.shp
Total buffer area: 8782.34 (CRS units squared)
True Positive: 1
True Negative: 4
False Positive: 0
False Negative: 2
Accuracy: 71.43%
Recall: 33.33%
```

```
Buffer geometries saved to SHP\IOT_BUFFER.shp
Total buffer area: 11918.88 (CRS units squared)
True Positive: 2
True Negative: 4
False Positive: 0
False Negative: 1
Accuracy: 85.71%
Recall: 66.67%
```

